

Safe Primes in the Eisenstein Integers

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Abstract

Safe primes, of the form $p = 2q + 1$ with q prime, occupy a privileged position in both number theory and cryptography. Their multiplicative group $\mathbb{F}_p^\times$ has order $2q$, yielding a large cyclic subgroup of prime order after factoring out the inevitable $\{\pm 1\}$ symmetry. A curious but forced arithmetic fact is that every safe prime $p > 3$ satisfies $p \equiv 2 \pmod{3}$: they automatically belong to the class of rational primes that remain irreducible (inert) in the Eisenstein integers $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$.

This observation naturally raises the question of whether the notion of “safety” can be formulated intrinsically within the Eisenstein setting. Since $\mathbb{Z}[\omega]$ has six roots of unity rather than two, the classical quotient $(p-1)/2$ suggests an Eisenstein analogue of the form $(N(\mathfrak{p})-1)/6$ for prime ideals $\mathfrak{p} \subset \mathbb{Z}[\omega]$.

We show that this leads to a sharp dichotomy. In the inert case $p \equiv 2 \pmod{3}$, the Eisenstein quotient $(p^2-1)/6$ is necessarily composite for all $p > 5$, ruling out a direct analogue of classical safety precisely in the congruence class where safe primes occur. By contrast, for split primes $p \equiv 1 \pmod{3}$, the quotient $(p-1)/6$ can be prime, giving rise to a robust and natural class of *Eisenstein-safe primes* of the form $p = 6q + 1$ with q prime.

Classical and Eisenstein safety therefore exhibit a striking complementarity: they occupy disjoint arithmetic progressions modulo 3, are governed by different unit groups ($\{\pm 1\}$ versus μ_6), yet arise from the same underlying group-theoretic template of isolating a large prime-order subgroup after factoring out unavoidable torsion.

We enumerate Eisenstein-safe primes, explore their density in

<https://safe-eisenstein-primes.vercel.app/> and discuss potential cryptographic applications for finite fields with hexagonal symmetry in

<https://leeleemacjames.github.io/safeTorsionCode/>.

Safe Primes and Why They Matter

Consider the prime $p = 23$. The nonzero elements modulo 23 form the multiplicative group

$$\mathbb{F}_{23}^\times = \{1, 2, \dots, 22\},$$

which is cyclic of order $22 = 2 \times 11$. Since 11 is prime, something remarkable happens: the squares modulo 23 form a subgroup

$$\begin{aligned} (\mathbb{F}_{23}^\times)^2 &= \{x^2 \bmod 23 : x \in \mathbb{F}_{23}^\times\}, \\ &= \{1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18\} \end{aligned}$$

of size exactly 11. This is a cyclic group of *prime order*. Because 11 is prime, this subgroup has no internal structure as every non-identity element generates the whole thing. From a group-theoretic perspective:

$$\mathbb{F}_{23}^\times \cong C_{22}, \quad (\mathbb{F}_{23}^\times)^2 \cong C_{11}.$$

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Clean exponent behavior

Fermat's Little Theorem tells us that for any $a \not\equiv 0 \pmod{23}$,

$$a^{22} \equiv 1 \pmod{23}.$$

Since $22 = 2 \cdot 11$, the only possible orders of elements are 1, 2, 11, or 22. This means raising any element to the power 11 produces a stark binary outcome:

$$a^{11} \equiv \pm 1 \pmod{23}.$$

In fact,

$$a^{11} \equiv 1 \pmod{23} \iff a \text{ is a square mod } 23,$$

while

$$a^{11} \equiv -1 \pmod{23} \iff a \text{ is a non-square mod } 23.$$

This simplicity of one large prime-order subgroup cleanly separated from a $\mathbb{Z}/2\mathbb{Z}$ symmetry, is what makes $p = 23$ special.

Definition: Safe primes

A prime p is called *safe* if

$$q = \frac{p-1}{2}$$

is also prime. Equivalently, $p = 2q + 1$ for some prime q . Complementing this we have the notion: a prime q is called a *Sophie Germain prime* if $p = 2q + 1$ is also prime. The first few safe primes are:

$5, 11, 23, 47, 59, 83, 107, 167, \dots$

Why cryptographers care

Many cryptographic protocols (Diffie–Hellman, DSA, etc.) prefer working in a cyclic group of large prime order. For a safe prime $p = 2q + 1$, one can work inside the subgroup of squares

$$G := (\mathbb{F}_p^\times)^2, \quad |G| = q \text{ (prime)}.$$

If g is a generator of $\mathbb{F}_p^\times$, then g^2 generates G . This is the arithmetic template we'll later try to reproduce in a different setting.

General formulation

For any safe prime p , we have $p - 1 = 2q$ with q prime. The unit group $\mathbb{F}_p^\times$ is cyclic of order $2q$, so it has exactly one subgroup of index 2: the quadratic residues. This subgroup has size q and is cyclic of prime order. The factorization $p - 1 = 2q$ forces clean exponent behavior. The order of any element $a \in \mathbb{F}_p^\times$ can only be 1, 2, q , or $2q$. Raising to the power q gives

$$a^q \equiv \pm 1 \pmod{p},$$

allowing membership in the square subgroup to be tested by a single exponentiation.

Classical safe primes originate in the arithmetic of $\mathbb{Z}$ [1], their geometric splitting behavior is governed by quadratic form theory [4], and their modern relevance is driven by cryptographic group structure [5].

The Eisenstein Congruence Class

Now observe something curious about the list of safe primes:

$$5, 11, 23, 47, 59, 83, 107, 167, \dots$$

Every safe prime greater than 3 satisfies

$$p \equiv 2 \pmod{3}.$$

Is this a coincidence? No, it's *forced*. If $p = 2q + 1$ with q an odd prime, then

$$p = 2q + 1 \equiv 2q + 1 \pmod{3}.$$

Since q is odd and greater than 2, we have $q \equiv 1$ or $2 \pmod{3}$.

- If $q \equiv 1 \pmod{3}$, then $p \equiv 2(1) + 1 = 3 \equiv 0 \pmod{3}$, so $p = 3$.
- If $q \equiv 2 \pmod{3}$, then $p \equiv 2(2) + 1 = 5 \equiv 2 \pmod{3}$.

Thus every safe prime $p > 3$ automatically lies in the arithmetic progression

$$p = 3k + 2.$$

But this congruence class $p \equiv 2 \pmod{3}$ has independent geometric and algebraic significance. These are precisely the primes that remain *irreducible* in the Eisenstein integers.

The Eisenstein integers

Define

$$\omega = e^{2\pi i/3} = \frac{-1 + \sqrt{-3}}{2}.$$

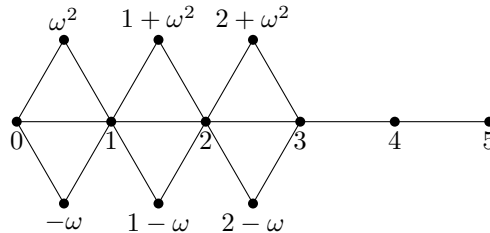
This is a primitive cube root of unity: $\omega^3 = 1$ and $\omega \neq 1$. It satisfies

$$\omega^2 + \omega + 1 = 0.$$

The *Eisenstein integers* are

$$\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}.$$

Geometrically, these form a triangular lattice in the complex plane:



Each vertex represents an Eisenstein integer, arranged in a hexagonal pattern with 120 angles instead of the 90 angles of the Gaussian integers $\mathbb{Z}[i]$. As an example of geometric visualization in the Eisenstein plane, see the author's work on the Eisenstein-Ulam spiral [7] and for a pedestrian visual introduction [6].

The norm form

The multiplicative norm is

$$N(a + b\omega) = (a + b\omega)(a + b\omega^2) = a^2 - ab + b^2.$$

This is always a non-negative integer, and

$$N(\alpha\beta) = N(\alpha)N(\beta).$$

For example:

$$N(2 - \omega) = 4 - 2(-1) + 1 = 7, \quad N(1 + \omega) = 1 - 1 + 1 = 1.$$

Splitting behavior of rational primes

Let p be a rational prime. When we pass to the Eisenstein integers, p can behave in three ways:

- **Ramified:** $p = 3$ factors as $(1 - \omega)^2$ up to units. The prime 3 is the only ramified prime.
- **Split:** If $p \equiv 1 \pmod{3}$, then p factors as $p = \pi\bar{\pi}$ where $\pi, \bar{\pi}$ are distinct Eisenstein primes of norm p . Equivalently, p is represented by the norm form:

$$p = a^2 - ab + b^2 \quad \text{for some } a, b \in \mathbb{Z}.$$

- **Inert:** If $p \equiv 2 \pmod{3}$, then p remains prime in $\mathbb{Z}[\omega]$. It is *not* represented by the norm form $a^2 - ab + b^2$.

Thus the congruence condition $p \equiv 2 \pmod{3}$ identifies those primes that stay irreducible. These are the *Eisenstein primes*.

Safe primes within the Eisenstein class

Among the Eisenstein primes (those with $p \equiv 2 \pmod{3}$) up to 53, we can identify which are safe:

p	$(p-1)/2$	Safe?
2	0.5	no
5	2	yes
11	5	yes
17	8	no
23	11	yes
29	14	no
41	20	no
47	23	yes
53	26	no

The safe primes among the Eisenstein primes up to 53 are:

$$\boxed{5, 11, 23, 47.}$$

This shows a key phenomenon:

- The Eisenstein condition $p \equiv 2 \pmod{3}$ selects a natural arithmetic class.
- Within this class, safe primes form a much thinner subset.
- Every safe prime > 3 automatically lies in the Eisenstein class, but most Eisenstein primes are *not* safe.

The Eisenstein congruence class provides the *ambient arithmetic setting*, while safety imposes a second, much stronger constraint.

Guiding question: Can the notion of “safety” be formulated *intrinsically* within the Eisenstein integers, rather than being inherited from $\mathbb{Z}$?

Why the Eisenstein Setting Is Special

Before addressing the guiding question, we need to understand why the Eisenstein integers occupy a privileged position among rings of algebraic integers.

The key is that three perspectives coincide here in a way that almost never happens:

1. The Eisenstein integers arise from a *cyclotomic field* $\mathbb{Q}(\zeta_3)$,
2. This cyclotomic field is itself *quadratic* (degree 2 over $\mathbb{Q}$),
3. Euler’s totient function $\varphi(3)$ simultaneously counts both units modulo 3 and Galois automorphisms.

Cyclotomic fields and Euler’s totient

For any integer $n \geq 1$, adjoining a primitive n th root of unity gives the cyclotomic field

$$\mathbb{Q}(\zeta_n),$$

which has degree

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n),$$

where $\varphi(n)$ is Euler’s totient function counting units modulo n . This equality reflects a canonical isomorphism

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

under which a unit $a \in (\mathbb{Z}/n\mathbb{Z})^\times$ acts by $\zeta_n \mapsto \zeta_n^a$.

Thus $\varphi(n)$ simultaneously counts:

- the number of units modulo n , and
- the number of field automorphisms.

This dual role is special to cyclotomic fields. Cyclotomic fields and their Galois structure are treated classically in [2].

When is a cyclotomic field quadratic?

A cyclotomic field is quadratic if and only if $\varphi(n) = 2$. This happens precisely for

$$n \in \{3, 4, 6\}.$$

The resulting fields are:

- $n = 3$: $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$ — the Eisenstein field,
- $n = 4$: $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i) = \mathbb{Q}(\sqrt{-1})$ — the Gaussian field,
- $n = 6$: $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\zeta_3)$ — the same as $n = 3$.

These are the *only* cyclotomic fields that are quadratic.

For the Eisenstein case $n = 3$:

$$\varphi(3) = 2 = |(\mathbb{Z}/3\mathbb{Z})^\times| = |\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})|.$$

The Galois group has two elements: the identity and complex conjugation $\omega \mapsto \omega^2 = \bar{\omega}$.

Contrast: the case $n = 5$

Consider $\mathbb{Q}(\zeta_5)$. Here

$$[\mathbb{Q}(\zeta_5) : \mathbb{Q}] = \varphi(5) = 4,$$

and

$$\text{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^\times \cong C_4.$$

This field is *not quadratic*. However, it contains a quadratic subfield: its maximal real subfield

$$\mathbb{Q}(\zeta_5)^+ = \mathbb{Q}(\zeta_5 + \zeta_5^{-1}) = \mathbb{Q}(\sqrt{5}).$$

Thus:

- $\mathbb{Q}(\sqrt{5})$ has degree 2, but
- it is only a *proper subfield* of the cyclotomic field,
- Euler's totient no longer directly measures its degree or arithmetic.

Splitting in $\mathbb{Q}(\sqrt{5})$ is controlled by the discriminant 5, not by the unit group modulo 5.

Summary: why $n = 3$ is special

For $n = 3$ (and $n = 4$), the cyclotomic field *is* quadratic. The ring of integers is:

$$\mathcal{O}_K = \mathbb{Z}[\omega], \quad K = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3}).$$

Key properties:

- Degree: $[K : \mathbb{Q}] = 2$,
- Discriminant: $\Delta_K = -3$,
- Unit group: $\mathcal{O}_K^\times = \mu_6 = \{\pm 1, \pm \omega, \pm \omega^2\}$ (the sixth roots of unity),

- Norm form: $N(a + b\omega) = a^2 - ab + b^2$.

Splitting behavior is clean:

- $p = 3$ ramifies,
- $p \equiv 1 \pmod{3}$ splits,
- $p \equiv 2 \pmod{3}$ is inert.

The Eisenstein integers occupy a sweet spot where cyclotomic, quadratic, and Galois-theoretic perspectives align perfectly.

Towards Eisenstein Safety

We now return to the guiding question: can we define “safety” intrinsically within $\mathbb{Z}[\omega]$?

In the present discussion, the word *torsion* will be used in a very mild and concrete sense.

Recall that $\mathbb{F}_p^\times$ is a finite cyclic group of order $p - 1$. An element $x \in \mathbb{F}_p^\times$ is said to have *finite order* if some positive power of x equals 1. In a finite group, every element has finite order, but some elements have *very small* order and are therefore unavoidable.

In $\mathbb{F}_p^\times$, the elements

$$1 \quad \text{and} \quad -1$$

satisfy

$$1^2 = 1, \quad (-1)^2 = 1,$$

so they are precisely the elements of order dividing 2. No matter which prime p we choose, these elements are always present.

For this reason, the set

$$\{\pm 1\} \subset \mathbb{F}_p^\times$$

is the smallest and most obvious “built-in symmetry” of the multiplicative group. It is a subgroup of size 2, and it cannot be removed or avoided.

When p is a safe prime, so that

$$p - 1 = 2q \quad \text{with } q \text{ prime,}$$

this subgroup accounts for the entire factor of 2 in the group order. Dividing the size of $\mathbb{F}_p^\times$ by 2 therefore removes exactly this inevitable symmetry:

$$\frac{|\mathbb{F}_p^\times|}{|\{\pm 1\}|} = \frac{p - 1}{2} = q.$$

Torsion as “what remains after dividing out the obvious symmetry”

From this point of view, the subgroup of squares

$$(\mathbb{F}_p^\times)^2$$

is simply what remains after identifying elements that differ by multiplication by ± 1 . Indeed,

$$\mathbb{F}_p^\times / (\mathbb{F}_p^\times)^2 \cong \{\pm 1\}.$$

Thus, in the safe prime case, the situation can be summarised as:

$$\mathbb{F}_p^\times \cong C_{2q}, \quad (\mathbb{F}_p^\times)^2 \cong C_q, \quad \mathbb{F}_p^\times / (\mathbb{F}_p^\times)^2 \cong C_2.$$

The phrase “torsion” here refers only to this unavoidable, small-order part of the group. That is, namely $\{\pm 1\}$ and the “safety quotient”

$$\frac{p-1}{2}$$

is simply the size of what remains once this obvious symmetry has been factored out. The reader is encourage to visit the app at <https://leelemacjames.github.io/safeTorsionCode/> to investigate this further.

The finite field perspective

For a safe prime p in $\mathbb{Z}$, the “obvious” torsion was $\{\pm 1\}$. In the Eisenstein setting, let $\mathfrak{p}$ be a nonzero prime ideal of $\mathbb{Z}[\omega]$. The quotient $\mathbb{Z}[\omega]/\mathfrak{p}$ is a finite field of size $N(\mathfrak{p})$, so

$$|(\mathbb{Z}[\omega]/\mathfrak{p})^\times| = N(\mathfrak{p}) - 1.$$

Crucially, $\mathbb{Z}[\omega]$ has *six* roots of unity:

$$\mu_6 = \{\pm 1, \pm \omega, \pm \omega^2\}, \quad |\mu_6| = 6.$$

These play the role that $\{\pm 1\}$ played in $\mathbb{Z}$. This suggests the Eisenstein safety quotient should be

$$\frac{N(\mathfrak{p}) - 1}{6}.$$

Remark (Ramification versus torsion). In the companion classification of Eisenstein–Pythagorean triples, [8] the ramified prime $\beta = 1 - \omega$ controls *primitivity*: a generator γ is “even” precisely when $\gamma \equiv 0 \pmod{\beta}$, and this vanishing forces a common factor 3 in the coordinates of γ^2 . In other words, primitivity is detected by the residue class of γ in the small quotient $\mathbb{Z}[\omega]/(\beta) \cong \mathbb{F}_3$. In this paper the same guiding idea reappears, but on the *multiplicative* side: every residue field $(\mathbb{Z}[\omega]/\mathfrak{p})^\times$ contains the unavoidable torsion subgroup of units μ_6 . Factoring out this built-in symmetry suggests the intrinsic “Eisenstein safety” quotient

$$\frac{|(\mathbb{Z}[\omega]/\mathfrak{p})^\times|}{|\mu_6|} = \frac{N(\mathfrak{p}) - 1}{6},$$

exactly paralleling the classical quotient $(p-1)/2$ obtained after removing $\{\pm 1\}$.

We will now return to the guiding question: can we define “safety” intrinsically within $\mathbb{Z}[\omega]$?

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so they are precisely the elements of order dividing 2. No matter which prime p we choose, these elements are always present. For this reason, the set

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this subgroup accounts for the entire factor of 2 in the group order. Dividing the size of $\mathbb{F}_p^\times$ by 2 therefore removes exactly this inevitable symmetry:

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Thus, in the safe prime case, the situation can be summarised as:

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The phrase “torsion” here refers only to this unavoidable, small-order part of the group which is $\{\pm 1\}$ and the “safety quotient”

$$\frac{p-1}{2}$$

is simply the size of what remains once this obvious symmetry has been factored out.

The Eisenstein analogue: a dichotomy emerges

In the Eisenstein setting, let $\mathfrak{p}$ be a nonzero prime ideal of $\mathbb{Z}[\omega]$. The quotient $\mathbb{Z}[\omega]/\mathfrak{p}$ is a finite field of size $N(\mathfrak{p})$, so

$$|(\mathbb{Z}[\omega]/\mathfrak{p})^\times| = N(\mathfrak{p}) - 1.$$

Crucially, $\mathbb{Z}[\omega]$ has *six* roots of unity:

$$\mu_6 = \{\pm 1, \pm \omega, \pm \omega^2\}, \quad |\mu_6| = 6.$$

These play the role that $\{\pm 1\}$ played in $\mathbb{Z}$. This suggests the Eisenstein safety quotient should be

$$\frac{N(\mathfrak{p}) - 1}{6}.$$

However, a striking dichotomy emerges depending on whether the underlying rational prime splits or remains inert:

- **Split case** ($p \equiv 1 \pmod{3}$): Here $p = \mathfrak{p} \cdot \bar{\mathfrak{p}}$ with $N(\mathfrak{p}) = p$, giving quotient $(p-1)/6$, which can be prime.
- **Inert case** ($p \equiv 2 \pmod{3}$): Here (p) remains prime with $N((p)) = p^2$, giving quotient $(p^2-1)/6 = (p-1)(p+1)/6$, which is *always composite* for $p > 5$.

This means that *Eisenstein-safe primes (in the strict sense of prime quotient) can only exist among split primes*, not among the inert primes where classical safe primes live.

Why the inert quotient is always composite

If $p \equiv 2 \pmod{3}$ and $p > 3$, then $p \equiv 5 \pmod{6}$, so $p+1 = 6m$ for some integer $m \geq 2$. Thus:

$$\frac{p^2-1}{6} = \frac{(p-1)(p+1)}{6} = (p-1) \cdot m.$$

For $p > 5$, both factors satisfy $m \geq 2$ and $p-1 \geq 6$, hence the product is composite.

Consequence: Classical safe primes (living in $p \equiv 2 \pmod{3}$) can *never* be Eisenstein-safe in the strict sense. The two notions of safety occupy *disjoint arithmetic progressions* modulo 3.

Eisenstein-safe primes: the split case

Since the inert case yields only composite quotients, we focus on split primes.

Definition. A split prime $p \equiv 1 \pmod{3}$ is *Eisenstein-safe* if

$$q := \frac{p-1}{6}$$

is prime. Equivalently, $p = 6q + 1$ with q prime.

When p splits as $(p) = \mathfrak{p} \cdot \bar{\mathfrak{p}}$, each factor has norm p , giving

$$|(\mathbb{Z}[\omega]/\mathfrak{p})^\times| = p-1 = 6q.$$

After factoring out the six roots of unity μ_6 , the remaining component has prime order q , exactly paralleling the classical safe prime template:

$$(\mathbb{Z}[\omega]/\mathfrak{p})^\times \twoheadrightarrow \mu_6, \quad \ker \cong C_q.$$

Examples of Eisenstein-safe (split) primes

The first few Eisenstein-safe primes are:

$$\boxed{13, 19, 31, 43, 67, 79, 103, 139, 223, 283, 307, 367, 439, 463, 523, \dots}$$

For instance:

- $p = 13 = 6 \cdot 2 + 1$, with $q = 2$ prime
- $p = 19 = 6 \cdot 3 + 1$, with $q = 3$ prime
- $p = 31 = 6 \cdot 5 + 1$, with $q = 5$ prime
- $p = 43 = 6 \cdot 7 + 1$, with $q = 7$ prime

Near 10^6 , examples include (see Appendix C):

$$997783, \quad 997807, \quad 998419, \quad 998743, \quad 999763,$$

for which $(p-1)/6 = 166297, 166301, 166403, 166457, 166627$ are all prime.

Up to 10^6 , there are 3201 Eisenstein-safe primes.

Contrast: classical safe primes cannot be Eisenstein-safe

Consider the classical safe prime $p = 23 = 2 \cdot 11 + 1$. Since $23 \equiv 2 \pmod{3}$, it is inert. The Eisenstein quotient is:

$$\frac{23^2 - 1}{6} = \frac{528}{6} = 88 = 8 \times 11,$$

which is composite. Similarly, for all classical safe primes $p > 3$:

- They satisfy $p \equiv 2 \pmod{3}$ (forced by $p = 2q + 1$)
- They are inert in $\mathbb{Z}[\omega]$
- Their Eisenstein quotient $(p^2 - 1)/6$ is always composite

The complementary nature of the two safety conditions

Safety type	Form	Congruence	Torsion quotient
Classical safe	$p = 2q + 1$	$p \equiv 2 \pmod{3}$ (inert)	$ \{\pm 1\} = 2$
Eisenstein-safe	$p = 6q + 1$	$p \equiv 1 \pmod{3}$ (split)	$ \mu_6 = 6$

The two notions exhibit a beautiful duality:

- They live in disjoint arithmetic progressions modulo 3
- They quotient out different natural torsion groups ($\{\pm 1\}$ vs μ_6)
- Both yield prime-order subgroups via the same group-theoretic template
- Neither implies the other: $p = 23$ is classically safe but not Eisenstein-safe; $p = 13$ is Eisenstein-safe but not classically safe

What about the inert case?

While inert primes cannot satisfy the strict condition (prime quotient), they may still be cryptographically useful if $(p^2 - 1)/6$ has a *large prime factor* with small cofactor.

For several classical safe primes $p \equiv 2 \pmod{3}$, define $E(p) := (p^2 - 1)/6$:

p	classical $(p-1)/2$	$E(p) = (p^2 - 1)/6$	factorisation of $E(p)$
47	23	368	$2^4 \cdot 23$
59	29	580	$2^2 \cdot 5 \cdot 29$
83	41	1148	$2^2 \cdot 7 \cdot 41$
107	53	1908	$2^2 \cdot 3^2 \cdot 53$
167	83	4648	$2^3 \cdot 7 \cdot 83$

Note that each $E(p)$ has the classical safe quotient $(p - 1)/2$ as a factor and this is automatic from $E(p) = (p - 1) \cdot (p + 1)/6$. While not prime-order, such primes may still offer cryptographic utility in $\mathbb{F}_{p^2}$ with hexagonal structure.

Remark (Complementarity with Eisenstein-Pythagorean Triples). The dichotomy between classical and Eisenstein safety reflects a deeper arithmetic complementarity. In the classification of Eisenstein-Pythagorean triples [8], the *Representability Principle* states that a prime p can appear as the “hypotenuse” c in a primitive solution to $a^2 - ab + b^2 = c^2$ if and only if p splits or ramifies in $\mathbb{Z}[\omega]$ —equivalently, $p = 3$ or $p \equiv 1 \pmod{3}$.

This creates a striking partition:

	Classical safe ($p = 2q + 1$)	Eisenstein-safe ($p = 6q + 1$)
Congruence	$p \equiv 2 \pmod{3}$	$p \equiv 1 \pmod{3}$
Behaviour in $\mathbb{Z}[\omega]$	Inert	Split
E-P hypotenuse?	No	Yes
Torsion quotient	$(p - 1)/2$	$(p - 1)/6$

Classical safe primes, being inert, are *forbidden* as hypotenuse values in the Eisenstein-Pythagorean equation. Conversely, Eisenstein-safe primes, being split, are *valid* hypotenuse values. The two notions of safety thus occupy complementary arithmetic territories, united by the same group-theoretic template (isolating a large prime-order subgroup after factoring out unavoidable torsion) but separated by the splitting/inert dichotomy.

Cryptographic take

In classical safe-prime cryptography one uses a prime $p = 2q + 1$ so that

$$|\mathbb{F}_p^\times| = p - 1 = 2q,$$

and then works in the unique subgroup of order q (prime). This prevents “small-subgroup” issues and yields a clean discrete logarithm group. In the Eisenstein split case, an Eisenstein-safe prime is

$$p = 6q + 1 \quad (q \text{ prime}).$$

Then

$$|\mathbb{F}_p^\times| = p - 1 = 6q.$$

So $\mathbb{F}_p^\times$ contains a unique subgroup of order q , and the remaining cofactor is just 6.

How one actually uses this. Choose a generator g of $\mathbb{F}_p^\times$ and define

$$h := g^6.$$

Then h lies in the subgroup of order q and (for a suitable choice of g) generates it. Equivalently, one checks

$$h^q \equiv 1 \pmod{p}, \quad h^{q/r} \not\equiv 1 \pmod{p} \text{ for every prime } r \mid q,$$

to ensure the order is exactly q . Thus Eisenstein-safe primes provide the same practical benefit as classical safe primes: a large prime-order group for discrete-logarithm protocols, with a small cofactor (6 instead of 2).

Remark. This is fundamentally about *group structure*, not integer factorization. Eisenstein-safe primes naturally suit discrete-logarithm protocols (Diffie–Hellman, DSA, elliptic curve variants) rather than RSA-type “composite modulus” schemes, [5]. The benefit is identical to classical safe primes: avoiding small-subgroup attacks by ensuring the working group has prime order.

Open Questions

The Eisenstein viewpoint produces two genuinely different “safety” regimes:

- **Split safety:** $p \equiv 1 \pmod{3}$ and $(p-1)/6$ prime.
- **Inert safety (relaxed):** $p \equiv 2 \pmod{3}$ and $(p^2-1)/6$ has a very large prime factor.

This suggests several questions.

1. **Counting and heuristics.** Let

$$\pi_E(x) := \#\{p \leq x : p \text{ prime and } (p-1)/6 \text{ prime}\}.$$

What heuristic constant C_E is suggested by the approximation

$$\pi_E(x) \sim C_E \frac{x}{(\log x)^2} ?$$

(Compare this to the usual heuristic for safe primes.)

2. **Comparative density.** Do Eisenstein-safe primes ($p = 6q + 1$) and classical safe primes ($p = 2q + 1$) have the same asymptotic density, or does the factor of 3 versus 2 create measurably different distributions? Empirically, up to 10^6 , there are 3201 Eisenstein-safe primes; how does this compare to classical safe primes in the same range?
3. **Efficient generation.** What is the fastest practical method to generate Eisenstein-safe split primes $p = 6q + 1$ with q prime, at cryptographic sizes, while also ensuring a usable generator of the order- q subgroup?
4. **Best possible inert analogue.** Since $(p^2-1)/6$ can never be prime for inert $p \equiv 2 \pmod{3}$, what is the right relaxed definition for cryptographic use? For example:

$$\frac{p^2-1}{6} = q \cdot s,$$

with q prime and s very small (or very smooth). How often does this occur, and what sizes of s are realistically attainable?

5. **A Gaussian comparison.** In the Gaussian case, the unit group has size 4. The split analogue would be primes $p \equiv 1 \pmod{4}$ with $(p-1)/4$ prime. How do the resulting “Gaussian-safe” and “Eisenstein-safe” populations compare in frequency and usefulness?
6. **Protocol-level meaning of hexagonal structure.** Beyond the congruence class $p \equiv 1 \pmod{3}$, is there any protocol-level advantage to representing field elements via Eisenstein coordinates in $\mathbb{Z}[\omega]/\mathfrak{p}$ rather than standard $\mathbb{F}_p$ arithmetic? Does the hexagonal lattice structure enable any novel cryptographic constructions, or is the benefit purely conceptual elegance?

Concluding Remarks

The interplay between classical and Eisenstein safety reveals a fundamental complementarity in arithmetic geometry. Where classical safe primes exploit the binary symmetry of $\{\pm 1\}$ within inert primes, Eisenstein-safe primes leverage the hexagonal symmetry of μ_6 within split primes. Neither generalizes the other; instead, they occupy disjoint arithmetic territories while sharing identical group-theoretic templates. This suggests that other cyclotomic settings, particularly the Gaussian integers with their fourfold symmetry, may harbor analogous "safety" notions.

A Cyclotomic Fields: A Broader View

The following table compares small cyclotomic moduli:

Modulus n	$\varphi(n)$	Quadratic?	Units $(\mathbb{Z}/n\mathbb{Z})^\times$	Inert primes	Notes
3	2	Yes	C_2	$p \equiv 2 \pmod{3}$	Eisenstein integers
4	2	Yes	C_2	$p \equiv 3 \pmod{4}$	Gaussian integers
5	4	No	C_4	$p \equiv 2, 3 \pmod{5}$	real quadratic subfield
6	2	Yes	C_2	$p \equiv 5 \pmod{6}$	same field as $n = 3$
7	6	No	C_6	$p \equiv 3, 5, 6 \pmod{7}$	primitive roots exist
8	4	No	$C_2 \times C_2$	—	non-cyclic units
9	6	No	C_6	$p \equiv 2, 5 \pmod{9}$	Eisenstein tower
12	4	No	$C_2 \times C_2$	—	square–hex fusion

Only $n \in \{3, 4, 6\}$ yield quadratic cyclotomic fields.

B Quadratic Fields

For comparison, here are selected quadratic fields (not all cyclotomic):

Discriminant Δ	Field	Ring of integers	Units	Inert primes	Origin
−3	$\mathbb{Q}(\sqrt{-3})$	$\mathbb{Z}[\omega]$	μ_6	$p \equiv 2 \pmod{3}$	cyclotomic
−4	$\mathbb{Q}(\sqrt{-1})$	$\mathbb{Z}[i]$	μ_4	$p \equiv 3 \pmod{4}$	cyclotomic
5	$\mathbb{Q}(\sqrt{5})$	$\mathbb{Z}[\frac{1+\sqrt{5}}{2}]$	infinite	$p \equiv 2, 3 \pmod{5}$	real subfield of $\mathbb{Q}(\zeta_5)$
−7	$\mathbb{Q}(\sqrt{-7})$	$\mathbb{Z}[\frac{1+\sqrt{-7}}{2}]$	$\{\pm 1\}$	$p \equiv 3, 5, 6 \pmod{7}$	Heegner
−8	$\mathbb{Q}(\sqrt{-2})$	$\mathbb{Z}[\sqrt{-2}]$	$\{\pm 1\}$	$p \equiv 5, 7 \pmod{8}$	quadratic only

Among imaginary quadratic fields, only $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$ arise as cyclotomic fields.

C Split case data table (up to 10^6)

In the split case $p \equiv 1 \pmod{3}$, the Eisenstein-safe condition is

$$q = \frac{p-1}{6} \text{ is prime} \quad (\text{equivalently } p = 6q + 1 \text{ with } q \text{ prime}).$$

For such a prime p , the splitting $(p) = \mathfrak{p} \bar{\mathfrak{p}}$ gives a residue field $\mathbb{Z}[\omega]/\mathfrak{p}$ of size p , hence

$$|(\mathbb{Z}[\omega]/\mathfrak{p})^\times| = p - 1 = 6q.$$

After factoring out the six always-present units $\{\pm 1, \pm\omega, \pm\omega^2\}$, one is left with a cyclic group of prime order q , exactly paralleling the classical safe-prime template.

Examples up to 500

p	$q = (p-1)/6$	check $p = 6q + 1$
13	2	$13 = 6 \cdot 2 + 1$
19	3	$19 = 6 \cdot 3 + 1$
31	5	$31 = 6 \cdot 5 + 1$
43	7	$43 = 6 \cdot 7 + 1$
67	11	$67 = 6 \cdot 11 + 1$
79	13	$79 = 6 \cdot 13 + 1$
103	17	$103 = 6 \cdot 17 + 1$
139	23	$139 = 6 \cdot 23 + 1$
223	37	$223 = 6 \cdot 37 + 1$
283	47	$283 = 6 \cdot 47 + 1$
367	61	$367 = 6 \cdot 61 + 1$
439	73	$439 = 6 \cdot 73 + 1$
499	83	$499 = 6 \cdot 83 + 1$

Large examples near 10^6

Up to 10^6 there are 3201 primes p satisfying this Eisenstein-safe split condition.

p	$q = (p-1)/6$	check $p = 6q + 1$
995587	165931	$995587 = 6 \cdot 165931 + 1$
996187	166031	$996187 = 6 \cdot 166031 + 1$
996487	166081	$996487 = 6 \cdot 166081 + 1$
996883	166147	$996883 = 6 \cdot 166147 + 1$
997099	166183	$997099 = 6 \cdot 166183 + 1$
997783	166297	$997783 = 6 \cdot 166297 + 1$
997807	166301	$997807 = 6 \cdot 166301 + 1$
998419	166403	$998419 = 6 \cdot 166403 + 1$
998743	166457	$998743 = 6 \cdot 166457 + 1$
999763	166627	$999763 = 6 \cdot 166627 + 1$

D Visualization of Classical safety on Eisenstein-Ulam spiral

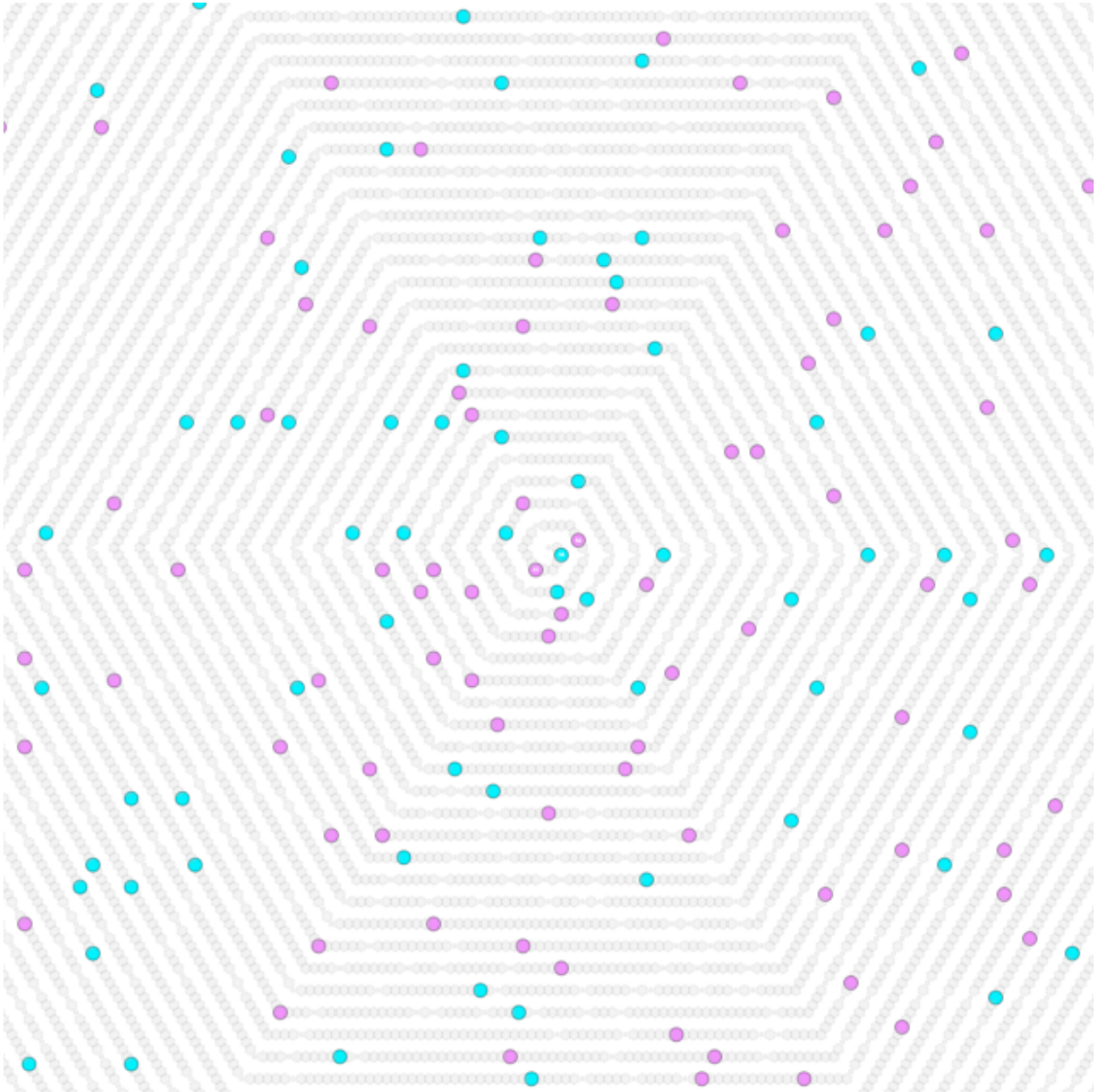


Figure 1: Safe primes mapped on an Eisenstein-Ulam spiral centered at 43

E Computational generation of Eisenstein-safe split primes

To enumerate Eisenstein-safe split primes up to a bound B , one proceeds as follows:

1. Loop over primes q with $q \leq (B - 1)/6$.
2. For each such q , form $p = 6q + 1$.
3. Test whether p is prime.
4. If so, record p as an Eisenstein-safe split prime.

No explicit congruence test modulo 3 is required, since $p = 6q + 1$ automatically satisfies $p \equiv 1 \pmod{3}$. The following Python code was used to generate all Eisenstein-safe split primes up to 10^6 :

```
from math import isqrt

def is_prime(n):
    if n < 2:
        return False
    if n % 2 == 0:
        return n == 2
    if n % 3 == 0:
        return n == 3
    r = isqrt(n)
    f = 5
    step = 2
    while f <= r:
        if n % f == 0:
            return False
        f += step
        step = 6 - step
    return True

def eisenstein_safe_split_primes(limit):
    out = []
    q_max = (limit - 1) // 6
    for q in range(2, q_max + 1):
        if is_prime(q):
            p = 6*q + 1
            if is_prime(p):
                out.append((p, q))
    return out

pairs = eisenstein_safe_split_primes(10**6)
print(len(pairs))
```

Interactive verification

The Eisenstein-Ulam spiral reveals striking ray patterns in prime distributions (McCulloch-James, 2025)[7]. Interactive tables of Eisenstein-safe primes are available online, developed by the author using Anthropic Claude:

`safe-classical-eisenstein.vercel.app` and `safe-eisenstein-primes.vercel.app`

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